

Setting up environment for Databases laboratories

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Zadanie 10 pn. „Przygotowanie i uruchomienie nowego kierunku studiów na studiach
II stopnia - Inżynieria i Analiza Danych (IAD)”
realizowane jest w ramach projektu
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środków Unii Europejskiej w ramach Europejskiego Funduszu Społecznego

Legal aspects

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- The terms of individual licenses can be different depending on which version of the software you select
- All legal constraints have to be strictly followed
- This presentation does not discuss legal terms of downloading and using the software, as these may be changed at any time by software vendors

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- Proszę dokładnie sprawdzić ograniczenia prawne, w tym warunki poszczególnych licencji dla produktów opisanych w niniejszej prezentacji
- Warunki poszczególnych licencji mogą się one różnić w zależności od wybranej wersji oprogramowania
- Wszystkie ograniczenia prawne muszą być ściśle przestrzegane
- Niniejsza prezentacja nie omawia prawnych warunków pobierania i używania oprogramowania, ponieważ mogą one zostać zmienione w dowolnym momencie przez dostawców oprogramowania

Ms SQL Server

Objectives

- Set up an environment to use it apart from and/or instead of laboratory machines
- Majority of the labs require just Ms SQL Server
- This presentation is focused on installing Ms SQL Server
- This can be done in one of two ways:
 - **Recommended option:** Ms SQL Server 2017 or 2019. Note that these releases are available both for Windows and Linux.
 - **The option requiring minimum hardware requirements:** Ms SQL Server 2008R2.
Note the some SQL SELECT queries will not run with this option. Moreover, data warehouses labs cannot be done with this edition. Otherwise, this option will be sufficient for majority of the lab tasks. This option is recommended in case no hardware matching Ms SQL Server 2017+ requirements is available.

Please note that during regular labs all the works are done on top of Ms Windows operating system. However, Linux users may decide to install their own private Ms SQL Server environment based on Linux.

Minimum hardware requirements

Minimum	Server® 2008 R2 SP2 - Express Edition	SQL Server 2017 Developer	SQL Server 2019 Developer
RAM	512 MB	1GB	1GB
OS	Windows XP, Windows Vista, Windows 7	Windows 8, Windows 10	Windows 10
Disk space	2.2GB	6GB	6GB
CPU	Pentium III-compatible processor AMD Opteron, AMD Athlon 64, Intel Xeon with Intel EM64T support, Intel Pentium IV with EM64T support	x64 Processor: AMD Opteron, AMD Athlon 64, Intel Xeon with Intel EM64T support, Intel Pentium IV with EM64T support	x64 Processor: AMD Opteron, AMD Athlon 64, Intel Xeon with Intel EM64T support, Intel Pentium IV with EM64T support

Compiled based on Ms SQL Server materials. Please note that comfortable work will require better hardware settings.

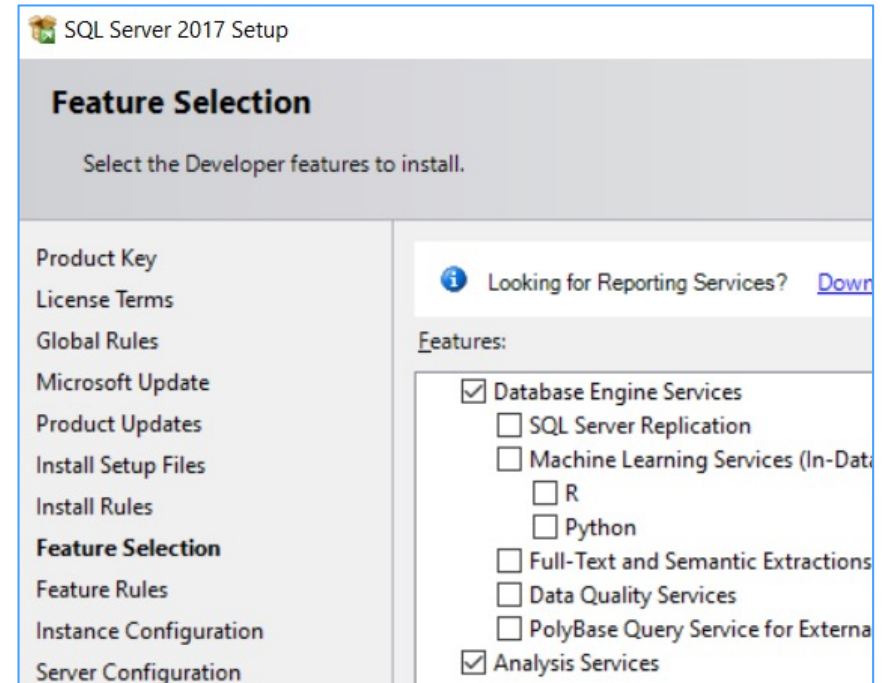
Downloading the software: SQL Server 2017 and 2019 for Windows

- Open <https://azureforeducation.microsoft.com/devtools>
- Select {Sign in} and log on with your ***@pw.edu.pl account
- Select {Free Software}/{Explore All}
- Select {SQL Server 2017 Developer} or {SQL Server 2019 Developer} and download it

Note that 2017+ releases are available also for Linux. See the final part of this presentation for details. In the case of macOS, please follow the Docker option of installation, also discussed at the end of this presentation

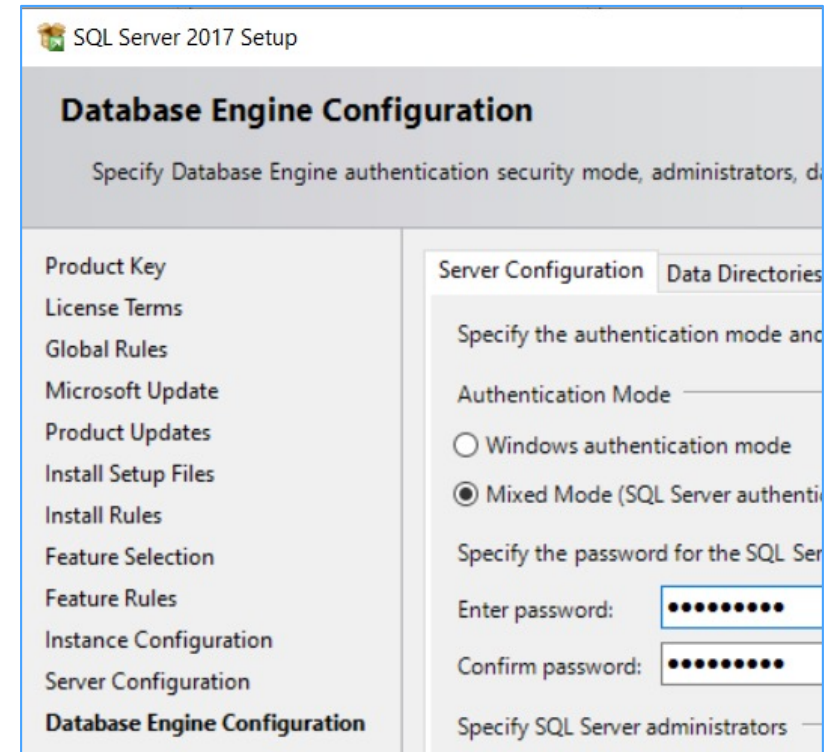
Installing the software: SQL Server 2017: part I

- run setup.exe from ISO installation media opened as a disk
- Select Installation/{New SQL Server stand-alone..}
- During {Feature Selection} stage, mark {Database Engine Services} and {Analysis Services}



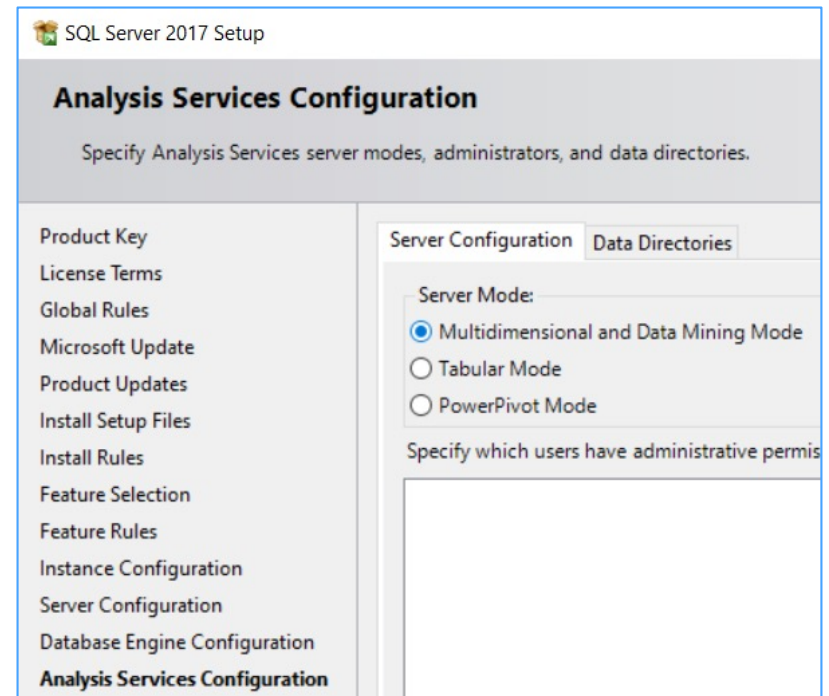
Installing the software: SQL Server 2017: part II

- Mark {Mixed mode (SQL Server authentication and (...))} and specify the password of sa user
- Add one of your Windows users as SQL Server administrator (typically a user with administrative privileges on a workstation). This can be done with {Add current user} or {Add...}



Installing the software: SQL Server 2017: part III

- During {Analysis Services Configuration} stage, mark {Multidimensional and data mining(...)}
- Add one of Windows users as Analysis Services administrator (typically a user with administrative privileges on the workstation). This can be done with {Add current user} or {Add...}
- Continue with remaining default settings and start installation



Installing the software: SQL Server 2017: part IV

- Now, from setup.exe:
 - Select Installation/{Install SQL Server Management Tools}
 - This will open a web browser and let you download SQL Server Management Studio (SSMS)
 - Next, SSMS should be installed
- After this stage, DBMS service, Analytical services (to be used for data warehouses lab), and SQL Server Management Studio are installed
- See next slide to learn how basic verification of your Ms SQL Server environment can be performed

Installing the software: SQL Server 2017: part V

- Start SQL Server Management Studio
 - select {SQL Server Authentication}
 - type in:
 - User: sa
 - Password: [PasswordDefinedDuringInstallation]
- If you connected, this means your SQL Server is up and running
- If not:
 - If your DBMS does not respond:
 - Check if it is properly installed and the service is started (by default it should be)
 - If DBMS responds, but user credentials are wrong, you may try to connect with {Windows authentication}

Installing Visual Studio with Microsoft Analysis Services Projects

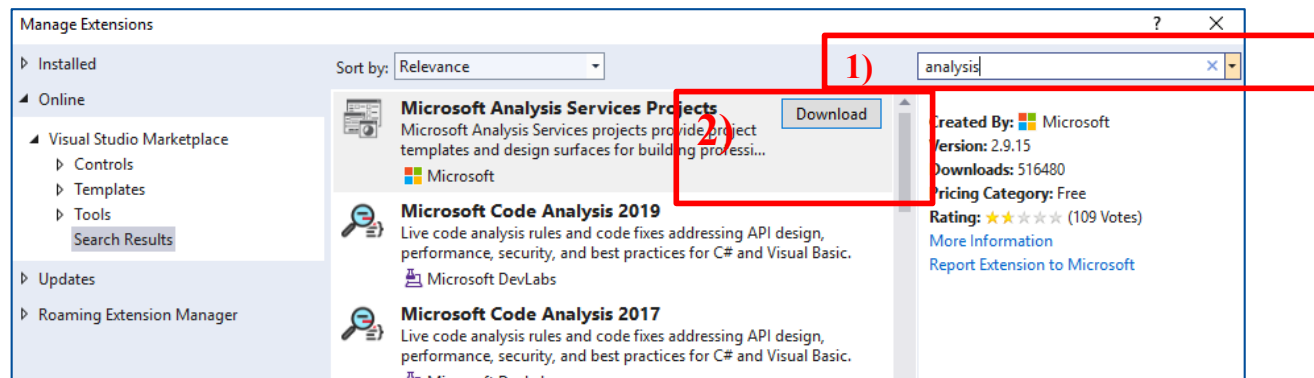
This part of presentation describes actions needed to prepare the environment for Business Intelligence projects that potentially will be done later during this semester

Installing the software: MS Visual Studio

- Note the Ms Visual Studio may be needed for the last regular scenario i.e. for laboratory 12 on the introduction to data warehouses
- In case you do not have it yet:
 - Open <https://azureforeducation.microsoft.com/devtools>
 - Select {Sign in} and log on with your ***@pw.edu.pl account
 - Go to {Free software}/{Explore All} and select Visual Studio to install Visual Studio 2019
 - During the installation, please select *Data storage and processing* (in Polish version: *Magazynowanie i przetwarzanie danych*) to be included

Installing *Microsoft Analysis Services Projects* extension I

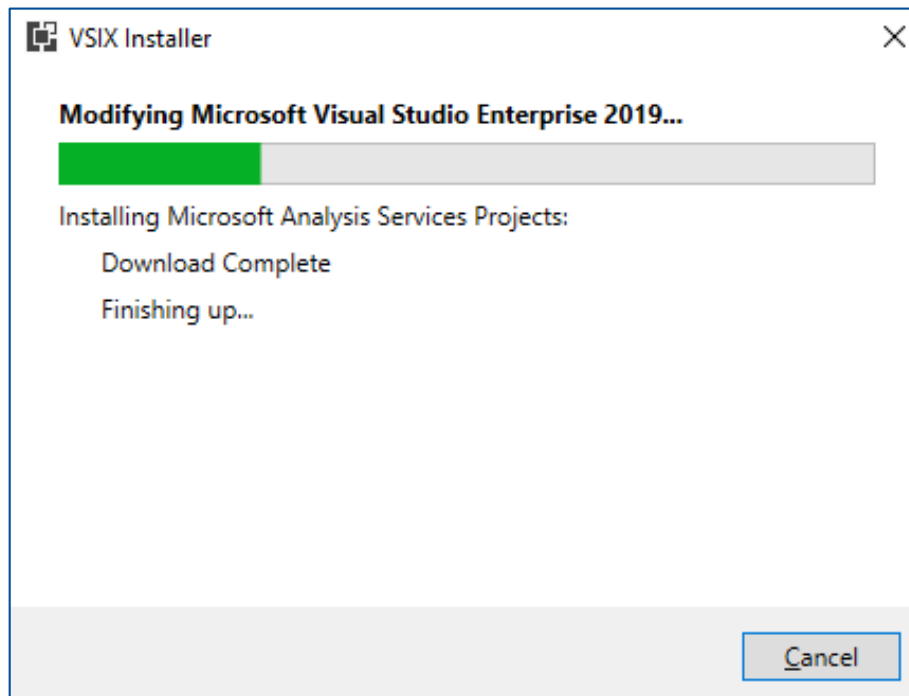
- Start Ms Visual Studio 2019 and:
 - In case you start it for the first time, select {Continue without code}
 - Select from main menu: {Extensions}/{Manage Extensions}, next {Visual Studio Marketplace}, and after finding {Microsoft Analysis Services Projects}, click on {Download}



- After downloading the software, close the window, which will start VSIX Installer, i.e. the installation of Microsoft Analysis Services Projects
- Click on {Modify} to confirm the installation of the software

Installing *Microsoft Analysis Services Projects* extension II

- This will start the setup of Visual Studio 2019 extension, which will enable the use of Analysis Services projects



- Once the installation process completes, Visual Studio can be used to run Analysis services projects

Installing alternative versions of Ms SQL Server

The remainder of this presentation is supposed to be used only when the installation of Ms SQL Server 2017+ on Ms Windows is not possible.

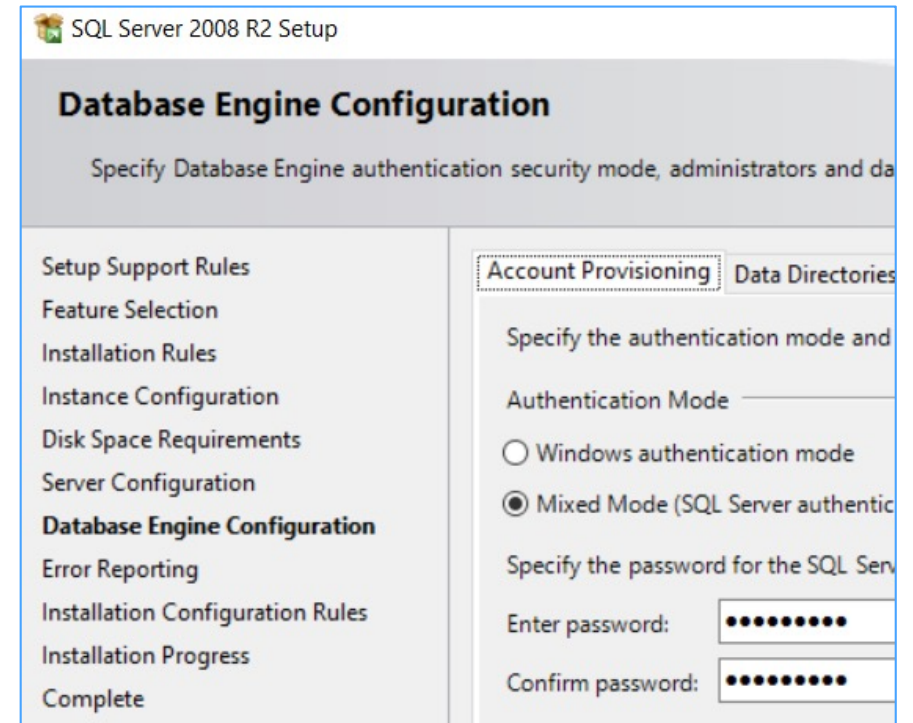
Downloading the software: SQL Server 2008 R2

- Open <https://www.microsoft.com/en-us/download/details.aspx?id=30438> to download Ms SQL Server 2008 R2 SP2
- Select SQLEXPRTWT_x64_ENU.exe
- Select {Installation/New Installation(...)}
- Note the limitations of SQL Server 2008 Express Edition:
 - it supports 1 physical processor, 1 GB memory, and 10 GB storage
 - **not all SQL queries will run on it**

This part is supposed to be used only when the installation of Ms SQL Server 2017+ is not possible.

Installing the software: SQL Server 2008: part I

- Mark {Mixed mode (SQL Server authentication and (...))} and specify the password of sa user
- Add one of your Windows users as SQL Server administrator, if needed (typically a user with administrative privileges on a workstation). This can be done with {Add current user} or {Add...}



Installing the software: SQL Server 2008: part II

- Start SQL Server Management Studio
 - select {SQL Server Authentication}
 - type in:
 - User: sa
 - Password: [PasswordDefinedDuringInstallation]
- If you connected, this means your SQL Server is up and running
- If not:
 - If your DBMS does not respond:
 - Check if it is properly installed and the service is started (by default it should be)
 - If DBMS responds, but user credentials are wrong, you may try to connect with {Windows authentication}

Ms SQL Server on Linux and macOS

- Please note that during regular labs, all the works are supposed to be done on top of Ms Windows operating system
- The other options are:
 - Ms SQL Server on Linux:
 - One may learn databases and run all laboratory exercises also with Ms SQL Server deployed on Linux
 - Linux users can consult <https://learn.microsoft.com/en-us/sql/linux/sql-server-linux-overview?view=sql-server-ver16> for details on installing SQL Server on Linux
 - Note that SQL Server 2017 is the first release that is supported on Linux. Prior versions were offered for Ms Windows only.
 - Ms SQL Server in Docker containers
 - See <https://learn.microsoft.com/en-us/sql/linux/sql-server-linux-docker-container-configure?view=sql-server-ver16&pivots=cs1-bash> for details, including sample dockerfiles
 - Note that this can be an option for MacOS users



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